CIVL 1100
Discovering Civil and Environmental Engineering

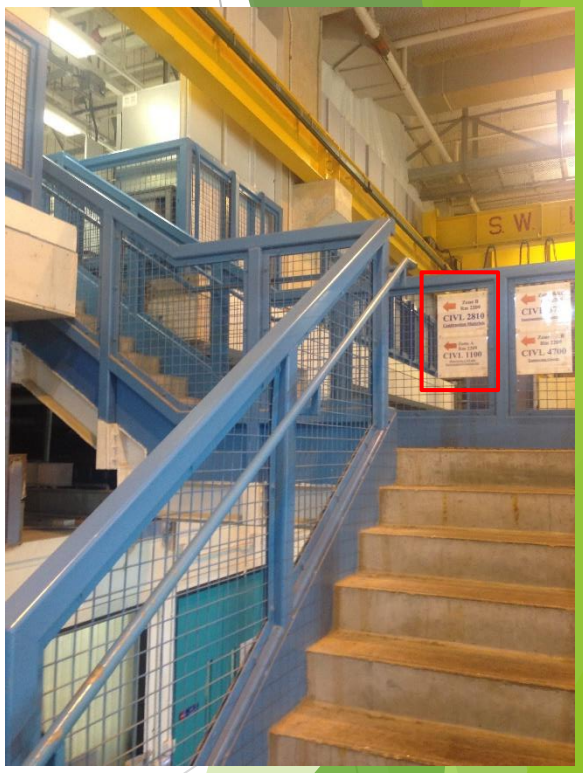
Introduction to Structural Engineering Experiments

Teaching Lab

Teaching Lab (cont'd)

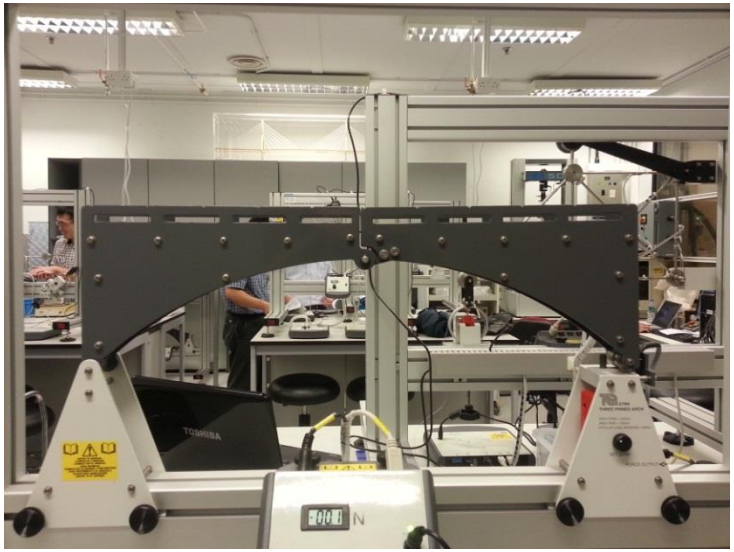
- Venue: Room 2209

Structures and materials laboratories (adjacent to lifts 19 and 20)

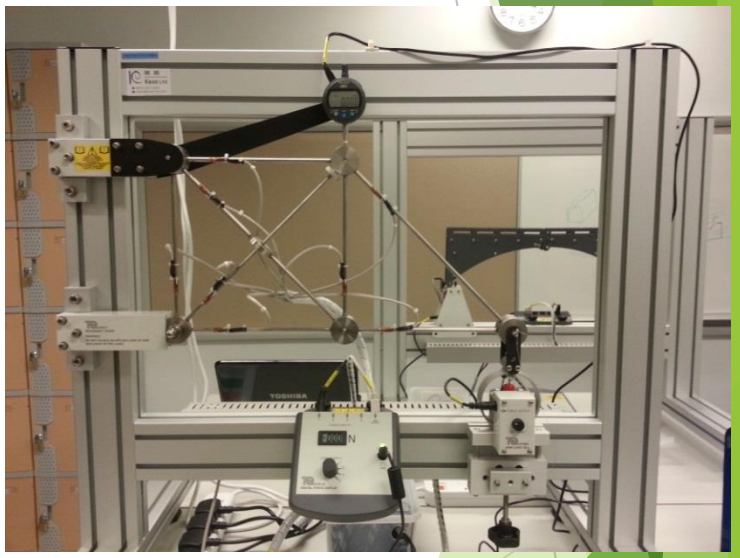


Introduction

- The lab session is composed of two tests:



Test 1 – Three-pinned Arch Experiment



Test 2 – Redundant Truss Experiment

Students' Grouping

Station 1 Truss

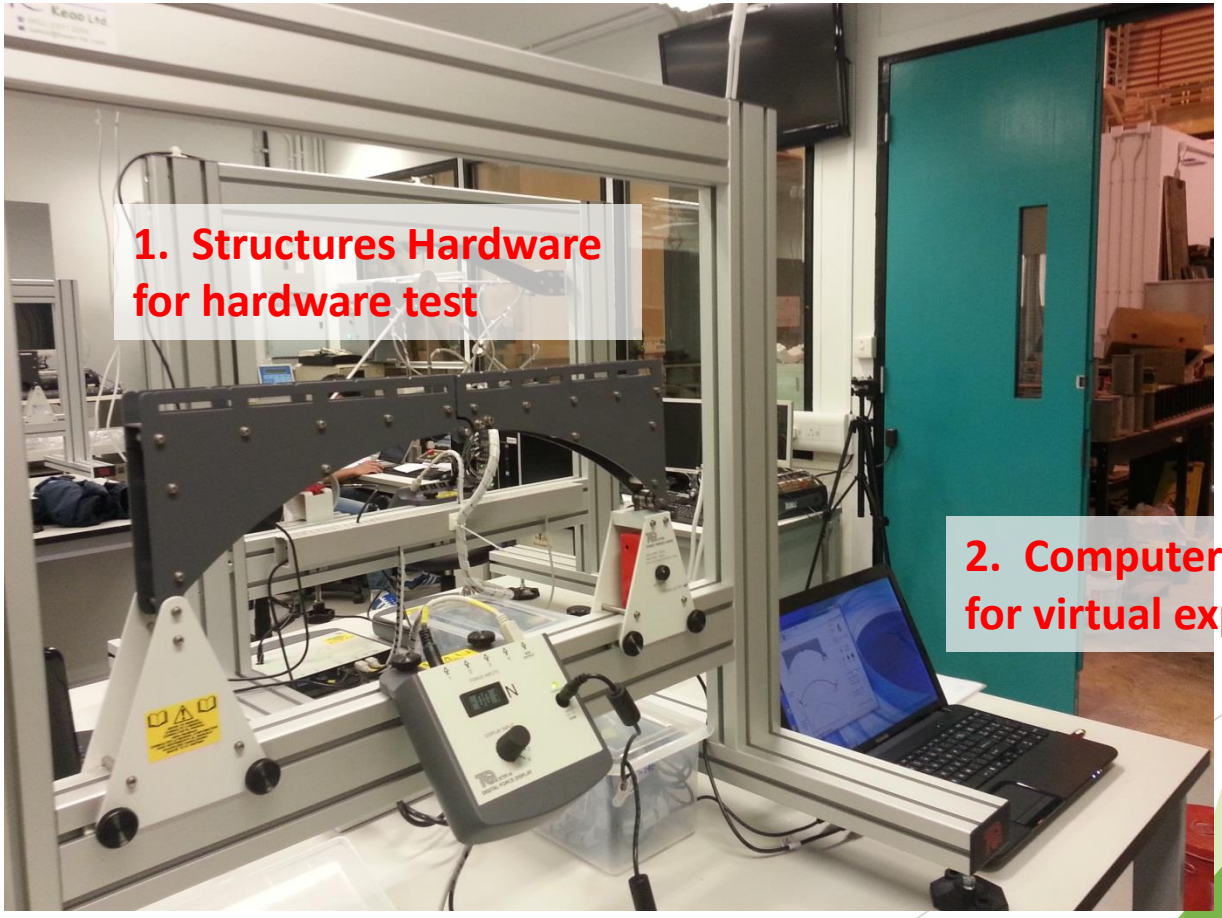
Station 2 Truss

Station 3 Arch

Station 4 Arch

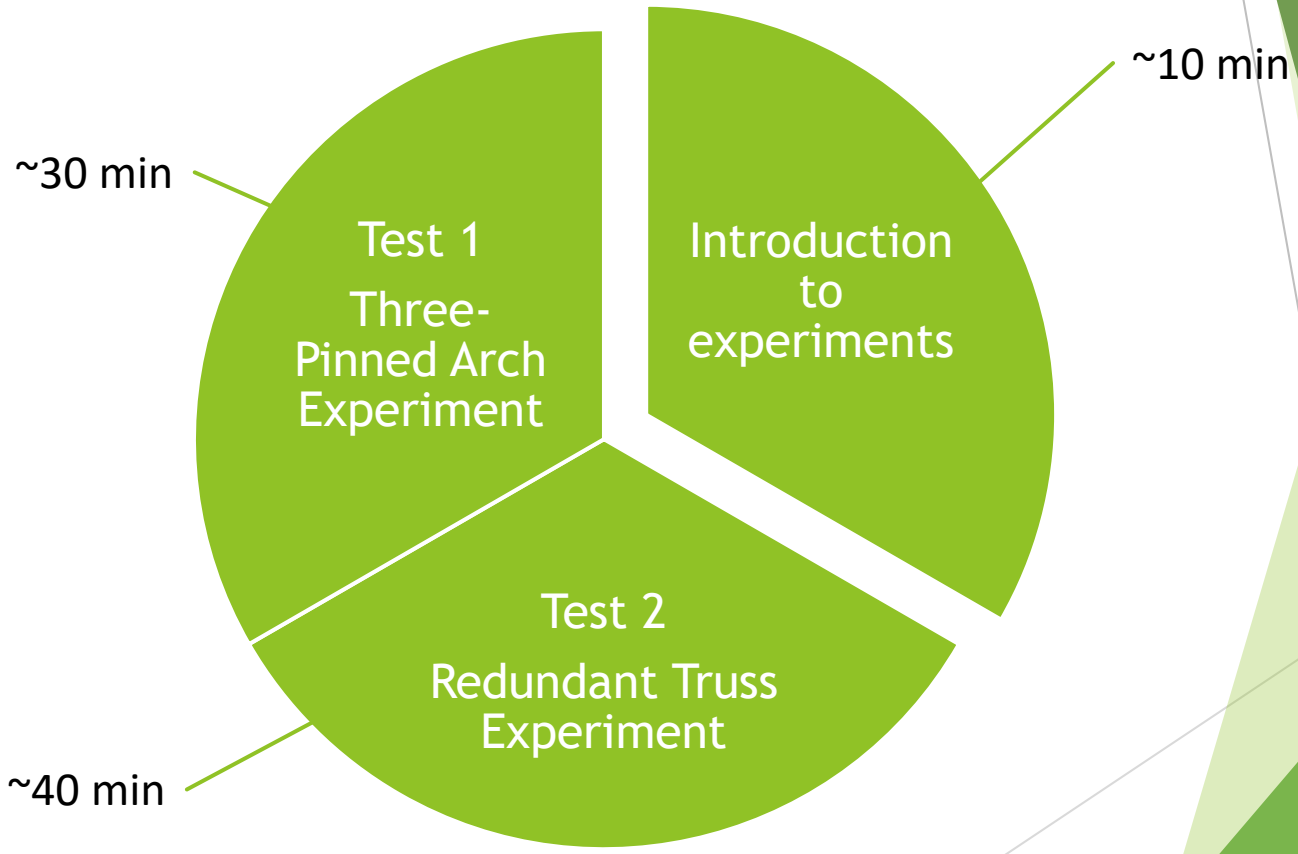
Introduction (cont'd)

- Each test (Test 1 and 2) is carried out using both



Time Schedule

Total time = 1 hour 20 mins

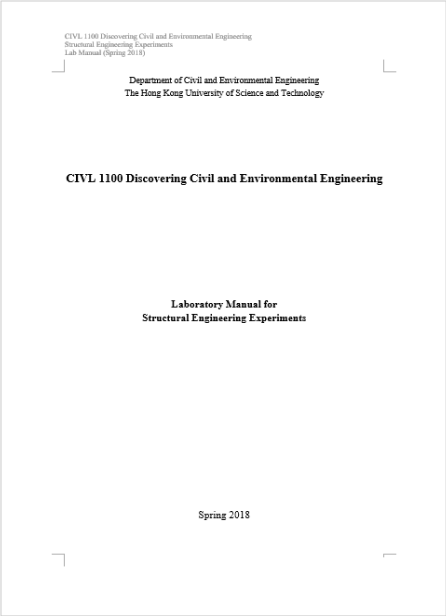


Objectives of this Experiment

- Gain hands-on experience in conducting hardware tests and virtual experiments.
- Have basic understanding of the structural behaviour of the two major structural forms in practice under vertical loading.
- Conduct simple calculations to analyse structures and verify theoretical formulas using experiment resul

Before the Experiment

- **Print out** the laboratory manual and **record sheets** (**at least one copy per group**).
- Please read the lab manual before the lab session.
- Make sure you know how to get to the teaching laboratory (Room 2209).



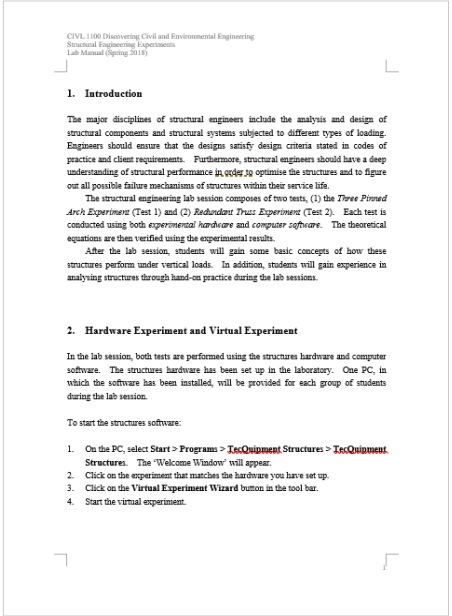
CIVL 1100 Discovering Civil and Environmental Engineering
Structural Engineering Experiments
Lab Manual (Spring 2018)

Department of Civil and Environmental Engineering
The Hong Kong University of Science and Technology

CIVL 1100 Discovering Civil and Environmental Engineering

Laboratory Manual for
Structural Engineering Experiments

Spring 2018



CIVL 1100 Discovering Civil and Environmental Engineering
Structural Engineering Experiments
Lab Manual (Spring 2018)

1. Introduction

The major disciplines of structural engineers include the analysis and design of structural components and structural systems subjected to different types of loading. Engineers should ensure that the designs satisfy design criteria stated in codes of practice and client requirements. Furthermore, structural engineers should have a deep understanding of structural performance in order to optimise the structures and to figure out all possible failure mechanisms of structures within their service life.

The structural engineering lab session composes of two tests, (1) the *Three Pinned Arch Experiment* (Test 1) and (2) *Redundant Truss Experiment* (Test 2). Each test is conducted using both *experimental hardware* and *computer software*. The theoretical equations are then verified using the experimental results.

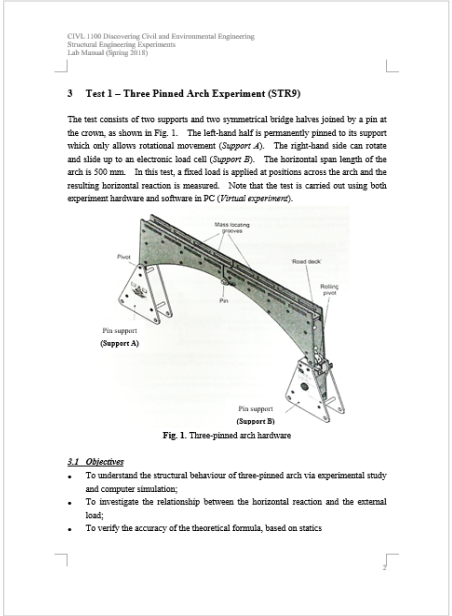
After the lab session, students will gain some basic concepts of how these structures perform under vertical loads. In addition, students will gain experience in analysing structures through hand-on practice during the lab sessions.

2. Hardware Experiment and Virtual Experiment

In the lab session, both tests are performed using the structures hardware and computer software. The structures hardware has been set up in the laboratory. One PC, in which the software has been installed, will be provided for each group of students during the lab session.

To start the structures software:

1. On the PC, select **Start > Programs > TecQuipment Structures > TecQuipment Structures**. The 'Welcome Window' will appear.
2. Click on the experiment that matches the hardware you have set up.
3. Click on the **Virtual Experiment Wizard** button in the tool bar.
4. Start the virtual experiment.



CIVL 1100 Discovering Civil and Environmental Engineering
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3 Test 1 – Three Pinned Arch Experiment (STR9)

The test consists of two supports and two symmetrical bridge halves joined by a pin at the crown, as shown in Fig. 1. The left-hand half is permanently pinned to its support which only allows rotational movement (*Support A*). The right-hand side can rotate and slide up to an electronic load cell (*Support B*). The horizontal span length of the arch is 500 mm. In this test, a fixed load is applied at positions across the arch and the resulting horizontal reaction is measured. Note that the test is carried out using both experiment hardware and software in PC (*Virtual experiment*).

Fig. 1. Three-pinned arch hardware

3.1 Objectives

- To understand the structural behaviour of three-pinned arch via experimental study and computer simulation;
- To investigate the relationship between the horizontal reaction and the external load;
- To verify the accuracy of the theoretical formula, based on statics

Basic Rules

- Come on time.
- Wear appropriate shoes and clothes. **No flip-flops!**
  - Your shoes should be able to protect your feet.
- No eating and drinking during the lab session.
- Please use the lab equipment **carefully**.
- Please follow the instructions by TAs and lab technicians.

Record Sheets

- You should record the data on the record sheets.
- **ONE completed set** of record sheets should be submitted with your lab report, or penalty will be given.

CIVL 1100 Record Sheets for Structural Engineering Experiments
Spring Term 2018

Table 1. Experimental data for Three Pinned Arch Test

*Percentage error = (Theoretical – Experimental) /

Mass	Distance from left	Span Fraction	Experimental horizontal reaction	Theoretical horizontal reaction	Theoretical vertical reaction (at support A)
g	mm	-	N	N	N

Table 2. Experimental data for Redundant Truss Test (**Statically Determinate** Structure)

Net strain (ε) = Measured strain at different force level - Measured strain when Force = 0 N (initial strain)

Negative net strain value represents compression

*Actual force (F) = Young's modulus (E) × Cross-sectional area of truss member (A) × **Net strain (ε)**

For Force = 0 N

Member	1	2	3	4	5	6	7	8
Measured strain ($\mu\varepsilon$)								
Theoretical strain ($\mu\varepsilon$)								

For Force = 50 N

Member	1	2	3	4	5	6	7	8
Measured strain ($\mu\varepsilon$)								
*Actual force (N)								
Theoretical strain ($\mu\varepsilon$)								
Theoretical force (N)								
Displacement (mm)	_____ (Hardware test)				_____ (Virtual experiment)			

For Force = 100 N

Member	1	2	3	4	5	6	7	8
Measured strain ($\mu\varepsilon$)								
*Actual force (N)								
Theoretical strain ($\mu\varepsilon$)								
Theoretical force (N)								
Displacement (mm)	_____ (Hardware test)				_____ (Virtual experiment)			

Lab Report Submission

Online through Canvas

Lab Report Submission

- The lab report is due **One Week** after your structural lab session.

Date of experiment	**27 February 2024**
Report due date	**5 March 2024**

Date of experiment	**5 March 2024**
Report due date	**12 March 2024**

Note:
You should submit the report at or before 11:59 pm on the due date.